นายอภิสิทธิ์ ธีรนะมงคลกุล 6609650228

①

1.1) $\Pi_{sname}(\sigma_{color = 'Green'}(Parts) \bowtie Catalog \bowtie Suppliers)$

1.2) $\Pi_{sid}(\sigma_{color = 'Red' \vee color = 'Black'}(Parts) \bowtie Catalog)$

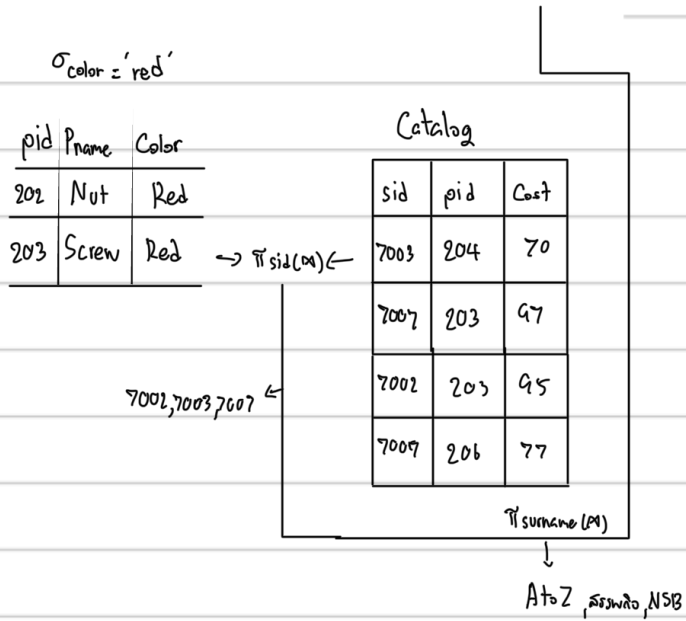
1.3) $\Pi_{sid}(Suppliers) - \Pi_{sid}(\sigma_{pid \neq pid}(Catalog))$

1.4) $\Pi_{pname}(\sigma_{cost > 200}(Catalog))$

1.5) $\Pi_{sid}(Suppliers) - \Pi_{sid}(Catalog)$

2.1.

$$\pi_{sname}(\pi_{sid}((\sigma_{color='red'} Parts) \bowtie (\sigma_{cost<100} Catalog)) \bowtie Suppliers)$$



A to Z, ascending, DSN

2.2.

7002, 7003, 7007

$$(\pi_{sid}((\sigma_{color='red'} Parts) \bowtie (\sigma_{cost<100} Catalog) \bowtie Suppliers)) \cap$$

$$(\pi_{sid}((\sigma_{color='green'} Parts) \bowtie (\sigma_{cost<100} Catalog) \bowtie Suppliers))$$

Pid	Pname	Color
206	Bolt	Green

$\pi_{sid}(Pg)$

7007

Sid	pid	Cost
7003	204	70
7007	203	97
7002	203	95
7007	206	77

so $\{7002, 7003, 7007\} \cap \{7007\} = 7007$

2.3.

$\rho(R1, Catalog)$
$\rho(R2, Catalog)$
$\pi_{R1.pid}\sigma_{R1.pid=R2.pid \wedge R1.sid \neq R2.sid}(R1 \times R2)$

pid = 202

sid	pid	cost
7002	202	255
7007	202	257
7005	202	252

pid = 203

sid	pid	cost
7007	203	97
7002	203	95

∴ pid = {202, 203}

③

2.1

```sql
SELECT DISTINCT S.sname
FROM Suppliers S
JOIN Catalog C ON S.sid = C.sid
JOIN Parts P ON C.pid = P.pid
WHERE P.color = 'red' AND C.cost < 100;
```

2.2

```sql
SELECT DISTINCT C1.sid
FROM Catalog C1
JOIN Parts P1 ON C1.pid = P1.pid
JOIN Suppliers S1 ON C1.sid = S1.sid
WHERE P1.color = 'red' AND C1.cost < 100

INTERSECT

SELECT DISTINCT C2.sid
FROM Catalog C2
JOIN Parts P2 ON C2.pid = P2.pid
JOIN Suppliers S2 ON C2.sid = S2.sid
WHERE P2.color = 'green' AND C2.cost < 100;
```

2.3

```sql
SELECT DISTINCT R1.pid
FROM Catalog R1, Catalog R2
WHERE R1.pid = R2.pid
AND R1.sid <> R2.sid;
```